

LAHIRUNI MALSHIKA

Software Engineer

📍 Matara, Sri Lanka
🌐 lahiruni-malshika

☎ +94 71 242 8170
🌐 lahiruni-malshika

✉ lahimalshi@gmail.com
🌐 lahiruni-malshika

PROFESSIONAL SUMMARY

Software Engineer with hands-on experience in building full-stack, AI-integrated web and mobile applications using the MERN stack and Flutter. Have a keen interest in cybersecurity and have skills in developing backend services, RESTful APIs, and creating responsive, accessible user interfaces with real-time data. Committed to producing clean, well-tested, and maintainable code, as well as participating in code reviews and following software engineering best practices. Eager to adopt new languages, frameworks, and tools to adapt quickly to evolving team needs.

WORK EXPERIENCE

Trainee Software Engineer at Virstack Pvt Ltd

DEC 2025 - AUG 2026

Tech Stack : MERN, Flutter/Dart, Docker, AWS, NGINX, CI/CD, WebSockets

- Contributed to developing and maintaining AI-powered full-stack applications using the MERN stack and cross-platform mobile development with Flutter, following coding standards.
- Optimized application performance for fast, reliable web and mobile experiences, integrated AI-driven features to enhance product functionality, and built scalable interfaces with clean software architecture.
- Supported DevOps workflows through Docker containerization and AWS cloud infrastructure, collaborating within an agile team using version control to deliver production-ready solutions.

Intern Software Engineer at Virstack Pvt Ltd

MAR 2025 - NOV 2025

Tech Stack : MERN, Angular, Flutter/Dart

- Contributed to full-stack development in web and mobile applications using React, Angular, Express with TypeScript, and Flutter (Dart), and conducted debugging and Api integrations for the backend.
- Optimized RESTful APIs and MongoDB aggregation pipelines for search, filtering, scheduling, and data-driven workflows, improving data retrieval efficiency and application responsiveness.

EDUCATION

BSc (Hons) in Information Technology - University of Moratuwa

JUN 2022 - AUG 2026

- Dean's List: L2S1 - 3.89, L2S2 - 3.94, L4S1 - 3.85, L4S2 - 3.88
- CGPA: 3.53 / 4.0

Diploma in English - BritishWay English Academy

FEB 2021 - MAR 2021

G.C.E. Advanced Level - Sujatha Vidyalaya Matara

JUN 2017 - OCT 2020

- Stream - Physical Science

TECHNICAL SKILLS

Programming Languages:	Java, C, Python, JavaScript, TypeScript, Dart
Mobile Development:	Flutter, Dart, State Management (Provider)
Frontend Development:	React.js, Next.js, JavaScript, Tailwind CSS, HTML, CSS
Backend Development:	Node.js, Express.js, Nest.js
Databases:	MySQL(NoSQL), MongoDB, Neo4j
Cloud & Deployment:	Docker, Kubernetes, CI/CD Pipelines, AWS S3/EC2/CloudWatch, NGINX
Testing & Quality Assurance:	Cypress, Selenium, Unit Testing, End-to-End Testing, Automation
Software Engineering Principles:	Git, OOP, Design Patterns, SOLID principles, REST APIs, Microservices Architecture, Agile/Scrum, AI coding assistants, Data Structures & Algorithms

PROJECTS

Final Year Research - Exploring Hardware-Software Co-design of Spatial Deep Learning Hardware Accelerators (Group project)

- As a Researcher, developed a cycle-accurate hardware-software co-design simulator for systolic array-based DNN accelerators to analyze and address performance bottlenecks in deep neural network execution.
- Designed and developed a Python-based compiler/mapper for workload mapping and memory management.
 - Role: Researcher
 - Technologies: Python, Verilog, DNN Workloads, FPGA Simulation
 - GitHub Repository

MozzaMelt - Ecommerce Platform (Group Project)

- An e-commerce pizza shop platform: contributed to building responsive and user-friendly interfaces.
- Implemented automated testing and supported quality assurance to improve application reliability, configuring and maintaining the CI/CD pipeline for streamlined development and deployment.
 - Role - Frontend Developer, Software Quality Assurance Engineer
 - Technologies - React JS, ASP.NET Core, MS SQL, Docker, CI/CD Pipeline, Selenium
 - Frontend GitHub Repository Backend GitHub Repository

Anothershots - Photography Focused Web Application (Level 2 Software Project - Mentored by 99X)

- Implemented the Featured Photos, Contact, and Packages sections of the photographer profile to enhance the portfolio.
- Designed and developed the Booking System and Admin Dashboard using clean architecture principles and reusable design patterns.
 - Role - Full Stack Developer
 - Technologies - Next JS, TypeScript, Tailwind CSS, NestJS, Prisma ORM, MongoDB
 - Frontend GitHub Repository Backend GitHub Repository

Baby Healthcare Mobile Application(Group Project)

- Developed features for a healthcare mobile application that connects users with medical providers and logs baby activities.
- Contributed to baby activity logs, activity history, timeline and its filters, and article modules using Flutter, Dart, and Provider state management, and conduct Unit and integration Testing.
 - Role - Mobile Application Developer, Backend Developer
 - Technologies - Flutter, Dart, Express Js, Node Js, MongoDB

Fuel Price Prediction Dashboard using Machine Learning (Individual Project)

- Built a machine learning dashboard to predict and visualize fuel price trends and prediction results in Sri Lanka using historical data.
- Performed data preprocessing, feature engineering, and model training to improve prediction accuracy.
 - Role: Machine Learning Engineer
 - Technologies: Python, Machine Learning, Pandas, Scikit-learn
 - GitHub Repository

Travel Recommendation Expert System (Individual project)

- Developed a rule-based expert system that recommends travel destinations using logical inference based on user preferences.
- Implemented features including best-season prediction and cheapest destination identification through Prolog rule evaluation.
 - Role: Machine Learning Engineer
 - Technologies: Python, Prolog (SWI-Prolog), Tkinter
 - GitHub Repository

SOFT SKILLS

- Problem Solving • Time Management • Leadership • Communication • Teamwork & collaboration

ACHIEVEMENTS AND PARTICIPATIONS

- **CyberZee'24 - Top 06** : An inter university quiz competition arranged by University of Kelaniya.
- **ACES Coders V11.0 2024 - Participant** : An inter university coding competition organized by University of Peradeniya.
- **MoraXtreme8.0 2023 - Top 75** : An inter university coding competition coordinated by IEEE student branch of University of Moratuwa.
- **Code Rush 2023 - Intra Faculty Coding Competition** : Participation
- **RealHack 5.0 - Participant** : An inter university coding competition organized by University of Kelaniya.

EXTRACURRICULAR ACTIVITIES

- **IEEE Student Branch of UoM** : Publicity committee member - Rise Up Mora 2023
- **IEEE Power & Energy Society Chapter of UoM** : WebMaster (Term of 24/25)
- **IEEE Women In Engineering Affinity Group of UoM** : Chief Organizer (Term 24/25), Marketing Committee Lead
- **Mathematics Society of UoM** : Publicity committee member - ENIGMA 2024
- **Leo Club of UoM**

REFERENCES

Dr. T.M. Thanthriwatta

Senior Lecturer,
Faculty of Information Technology,
University of Moratuwa.
Tel: +94 71 815 2337
Email: thilinat@uom.lk

Mr. Nimna Perera

Senior Software Engineer
99X.io
Tel: +94 71 015 5336
Email: nimnap@99x.io